Mechanical Engineering

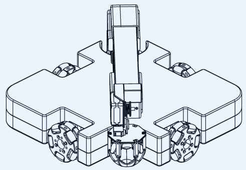
Portfolio

Hi, I’m **Shourya Sahdev!**

A Mechanical Engineer based in Illinois, USA, passionate about building reliable, functional, and manufacturable products.
I specialize in CAD, FEA, and rapid prototyping—turning concepts into high-performance, real-world solutions.

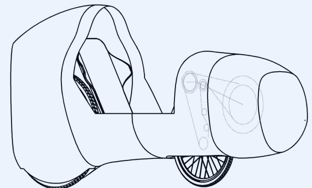
Mechanical Design | Product Development | Simulation & Testing

Projects



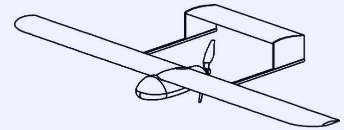
01 Explore >

Mobile Robotic Arm
- Kinematics
- DFMA
- System Integration



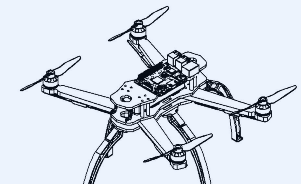
02 Explore >

Recumbent Bicycle
- Composites
- Design Validation
- Material Selection



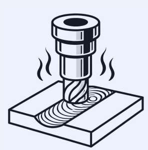
03 Explore >

Fixed Wing UAV
- CFD
- Flight Mechanics
- Multidisciplinary Des.



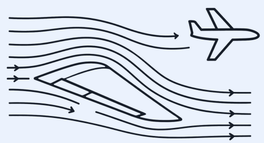
04 Explore >

Quadcopter
- Design Optimization
- Structural Analysis
- Additive Mfg.



05 Explore >

FSP Parameter Study
- FEA
- Thermal Analysis
- Process Simulation

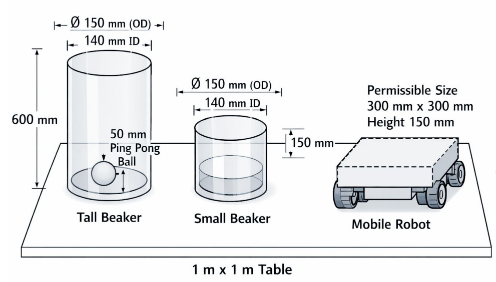


06 Explore >

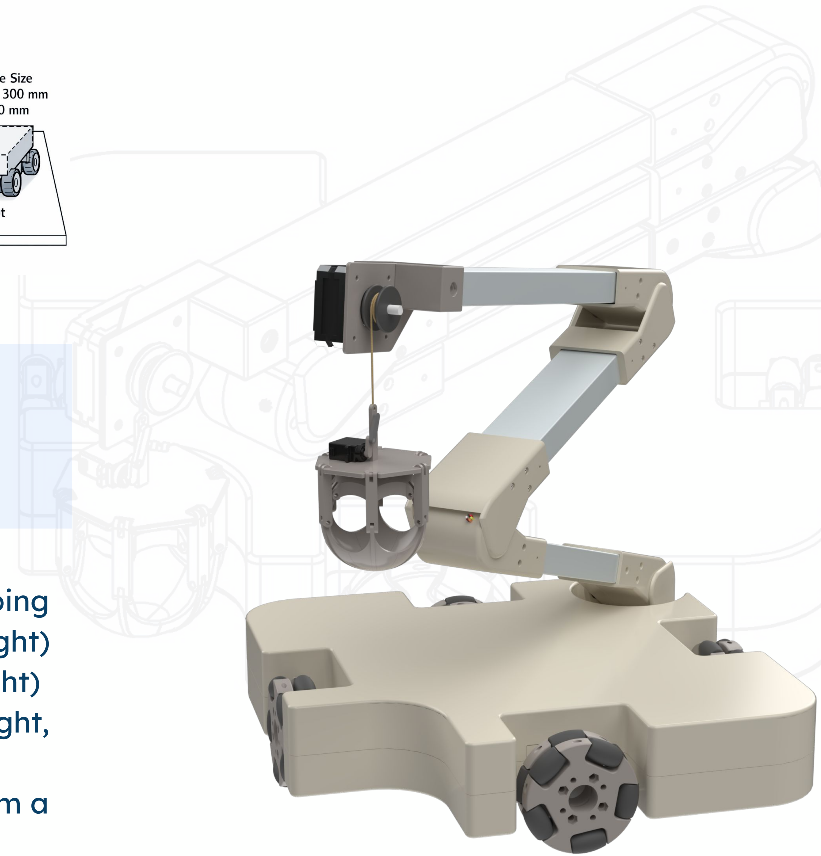
Impact of Wing Sweep
- Numerical Methods
- Aerodynamic Modeling
- Python

07 Explore >

Beam Impact Test
- Data Acquisition
- Experimental Mechanics
- Structural Testing



01 Mobile Robotic System for Precision Object Retrieval

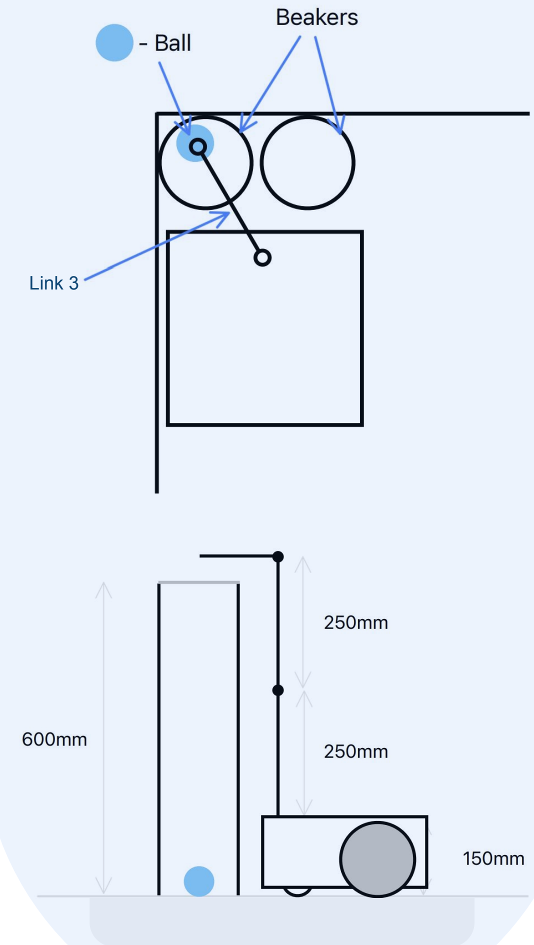


- Designed a mobile robotic system to extract a ping pong ball from a tall glass beaker (600 mm height) and transfer it into a smaller beaker (150 mm height)
- Required compliance with strict dimensional, weight, and mobility constraints
- System designed for remote human operation from a distance of 10m with repeatable task execution

Mobile Robotic Arm

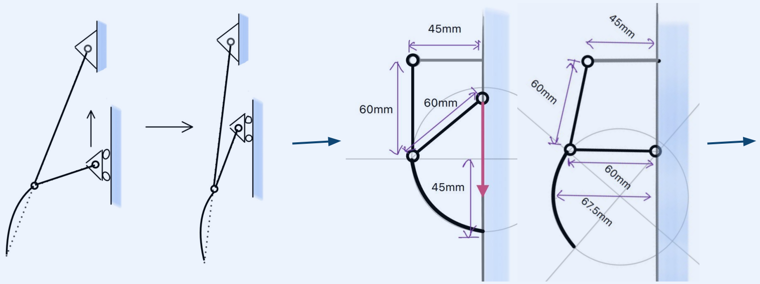
Design Challenges & Edge Cases

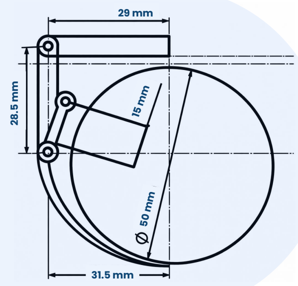
- Identified and designed around the worst-case configuration to guarantee performance across all scenarios
- Edge case defined as:
  - Tall beaker placed at the table corner
  - Ping-pong ball located at the furthest internal edge and maximum depth
  - Second beaker positioned immediately adjacent to the tall beaker
- This configuration imposed simultaneous requirem-
-ents on:
  - Maximum horizontal and vertical reach
  - High positional precision near fragile containers
  - Controlled approach angles to avoid contact or toppling



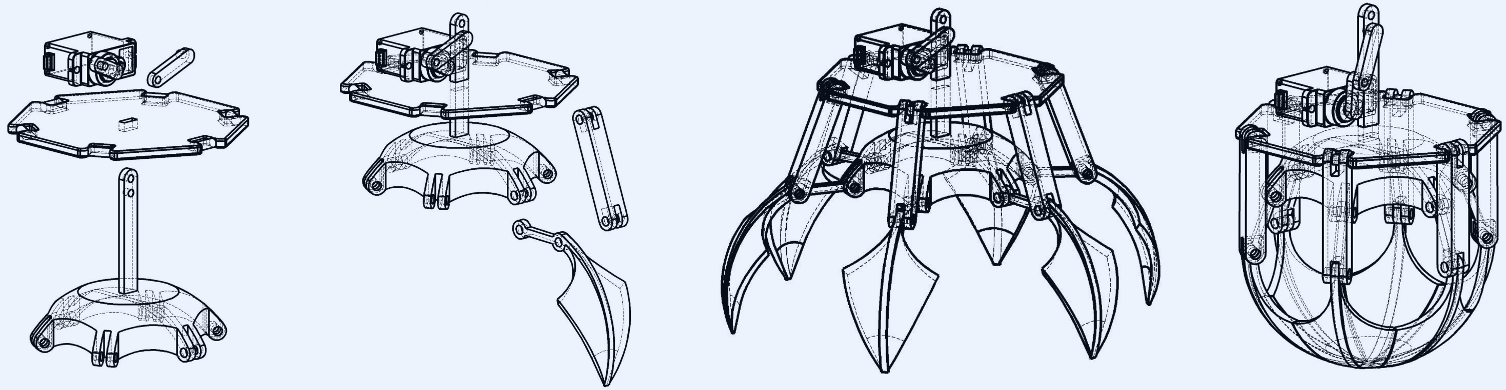
Mobile Robotic Arm

End Effector Design Strategy



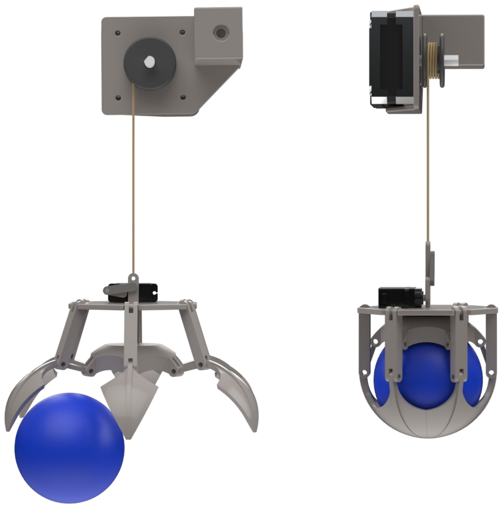


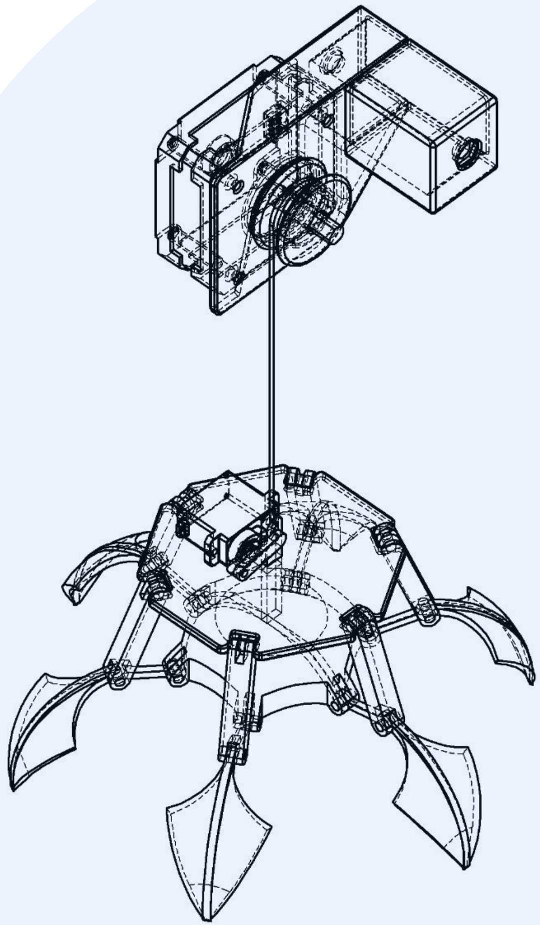
- Evaluated multiple extraction conc--epts including grippers, suction, and magnetic approaches
- Designed a 3-link gripper geometry sized for a 50 mm spherical object
- Enabled secure capture by sliding under the ball rather than pinching



Passive Compliance & Reliability

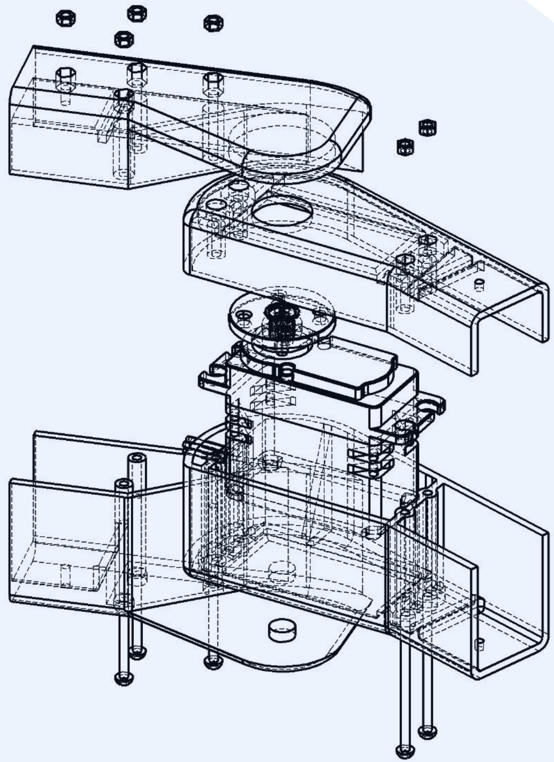
- Implemented a thread-based lowering mechan--ism to introduce passive compliance
- Allowed self-alignment with beaker walls during descent
- Reduced operator accuracy requirements and improved real-world robustness

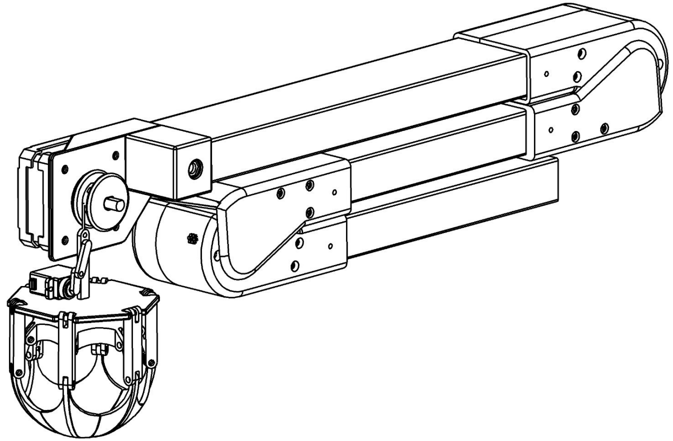




Mobile Robotic Arm

Manipulator Architecture & Joints



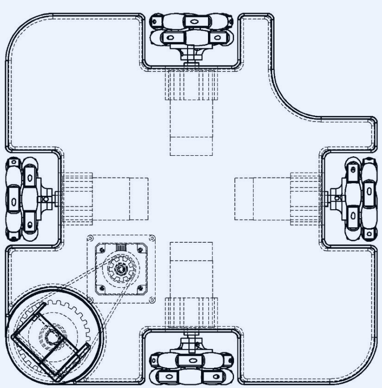


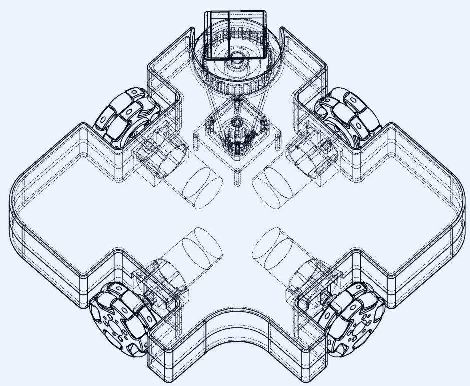
- Selected a three-link articulated arm for vertical reach and precise positioning
- Designed servo-driven joints with 180° rotation and compact folding
- Standardized joint geometry to reduce part count and simplify manufacturing

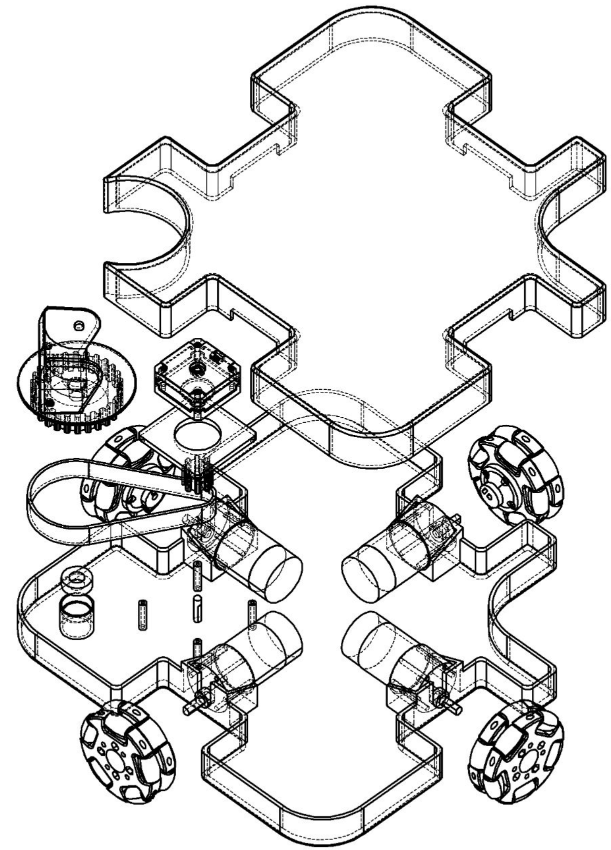
Mobile Robotic Arm

Mobile Platform Selection

- Compared differential, tracked, omni-wheel, and Mecanum drive systems
- Selected a four omni-wheel platform for precision translation and rotation
- Enabled 360° rotation even when operating next to obstacles



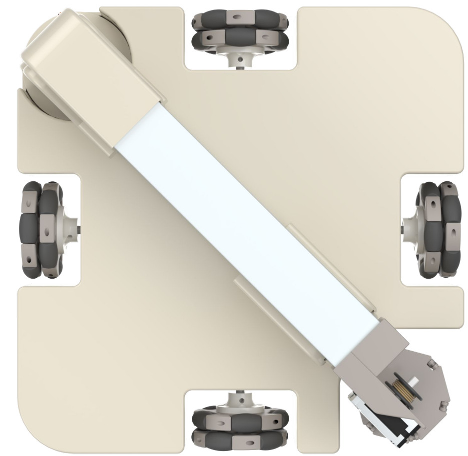


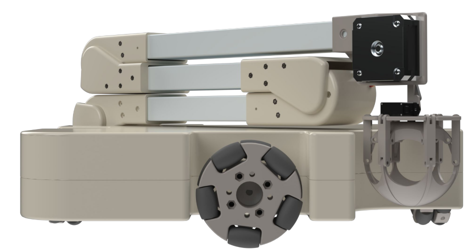


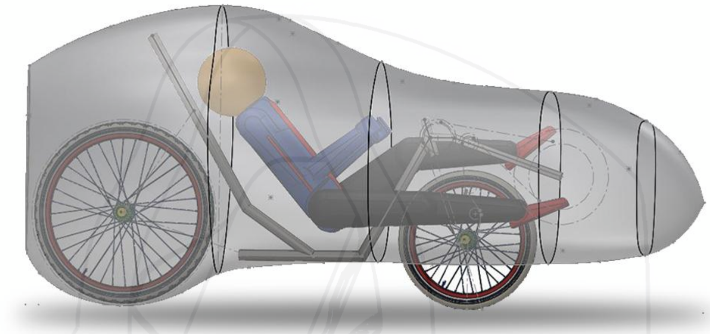
Mobile Robotic Arm

Evaluation & Engineering Takeaways

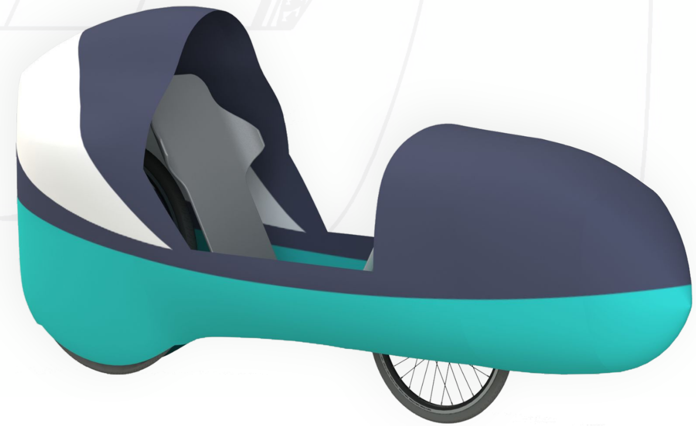
- Achieved a compact 300 × 300 mm footprint with a total mass of ~3.66 kg
- Used SolidWorks for CAD-driven design, mass estimation, and assembly integration
- Demonstrated system-level robotics thinking, DFM, and mechanism design skills







02 Design & Development of a Fully Faired Recumbent HPV



- Designed a fully faired recumbent human powered vehicle for ASME HPVC 2020
- Focused on ergonomics, handling, drivetrain efficiency, and aerodynamic performance
- Developed two vehicle iterations over two years with continuous design improvements

Vehicle Architecture & Configuration

- Selected fully faired low-racer recumbent configuration for aerodynamic advantage
- Reduced wheelbase and tightened seating position to improve maneuverability
- Front wheel drive layout chosen for compact packaging and power transmission

Frame Design & Stability Analysis

- Designed frame using AISI 4130 Chromoly steel for strength-to-weight balance
- Conducted parameter study using Whipple model for stability optimization
- Finalized geometry based on head tube angle, trail, wheelbase, and wheel size

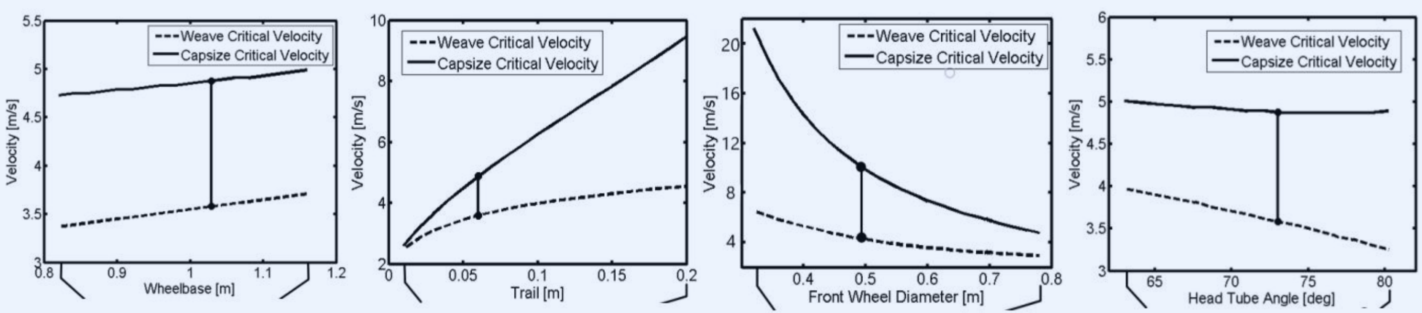
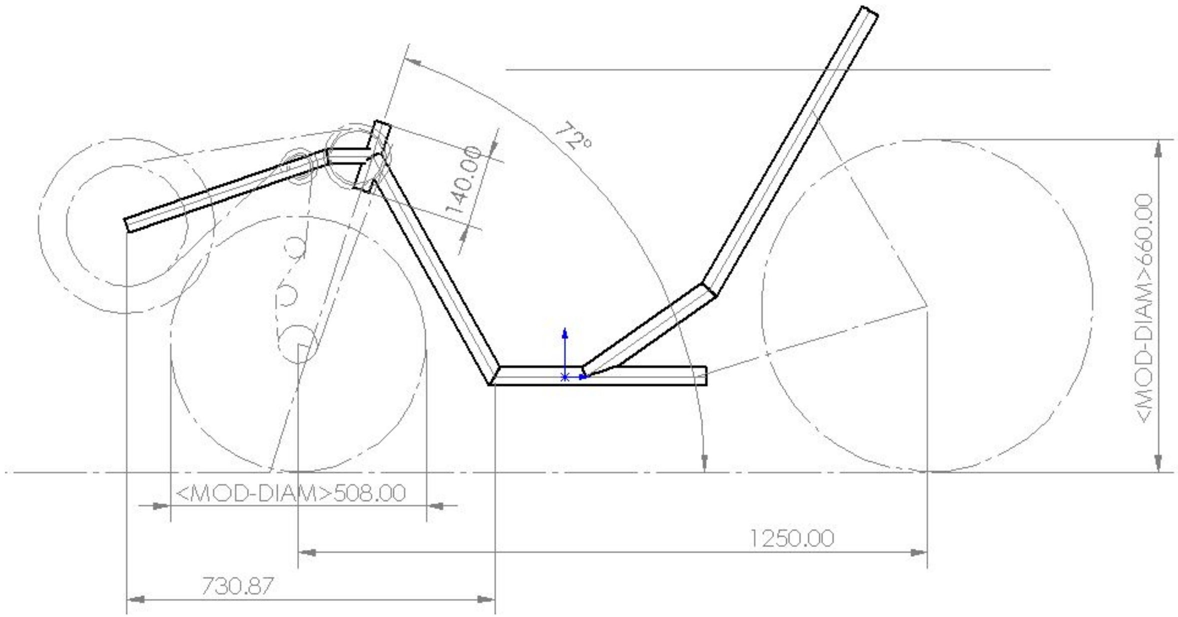


Figure: Whipple model

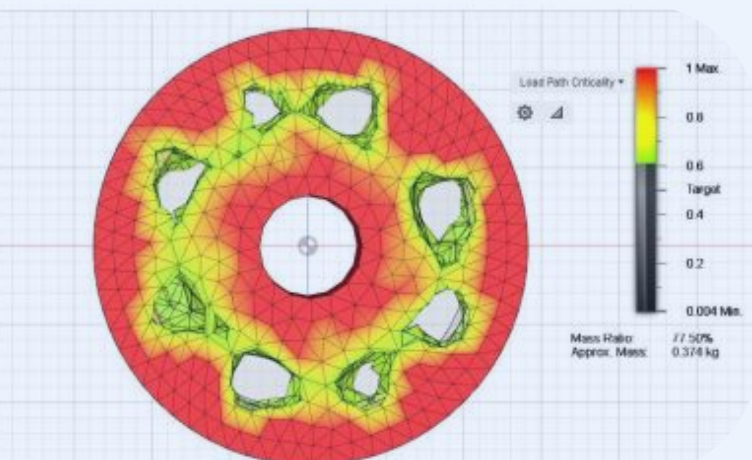
Ergonomics & Rider Power Output

- Optimized rider posture to maximize sustained power output and comfort
- Positioned bottom bracket and chainring to improve pedaling efficiency
- Evaluated turning capability with chainline and knee clearance considerations



Structural Optimization & Drivetrain Design

- Performed topology optimization using human pedaling load cases
- Designed idler system for efficient routing, tension relief, and derailment prevention
- Optimized drivetrain components to balance efficiency, reliability, and packaging



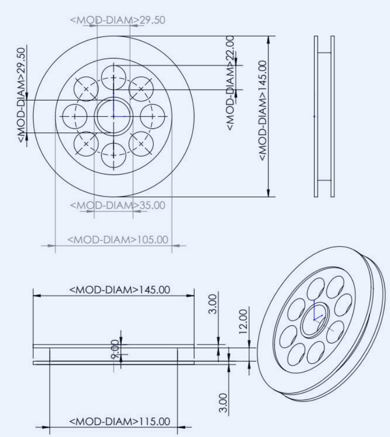


Figure: Idler Topology Result using Fusion 360

Aerodynamic Fairing Design & Development

- Developed fairing geometry based on rider heel, toe, and knee motion envelopes
- Iterated fairing shape using airfoil sections and computational airflow studies
- Selected final teardrop-canopy form to minimize drag while maintaining clearance

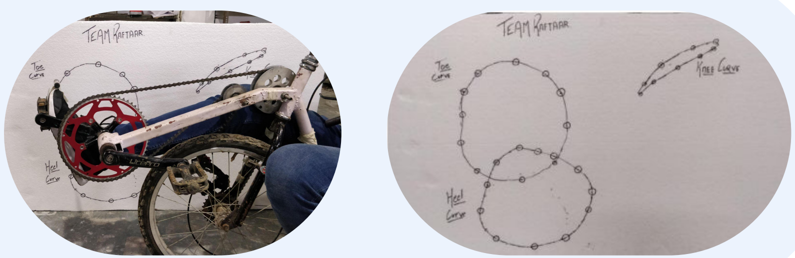
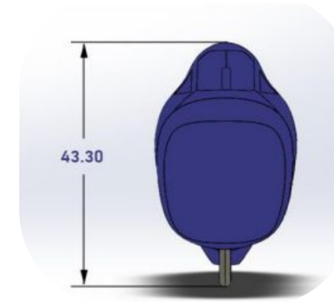
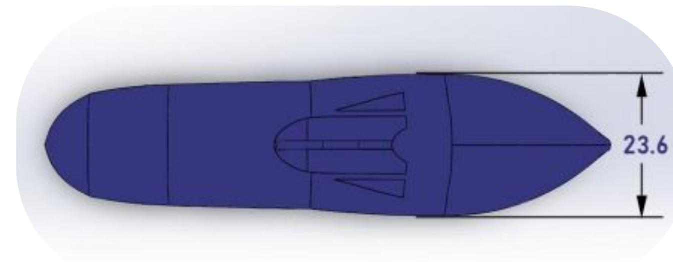
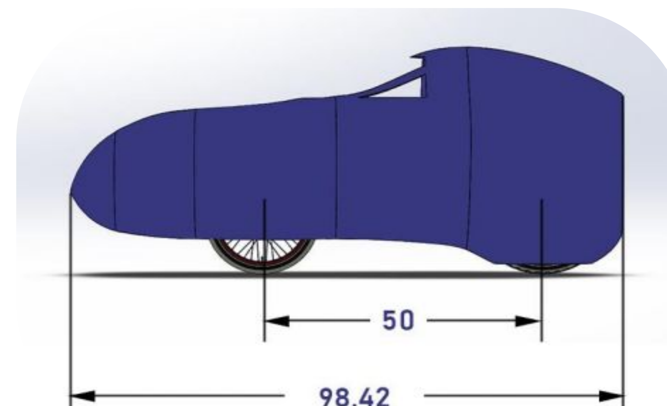


Figure: Measurement of Heel min, Toe max and Knee Curves







Design & Development of a Fully Faired Recumbent HPV

CFD Analysis & Structural Validation

- Conducted steady-state CFD using k-omega turbulence model in Ansys Fluent
- Ensured mesh independence through progressive refinement and convergence checks
- Validated fairing strength through top and side load testing with high safety factors

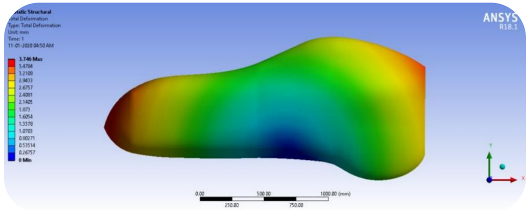


Figure: Top Load Testing Results

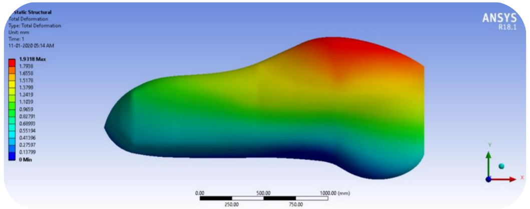


Figure: Side Load Testing Results

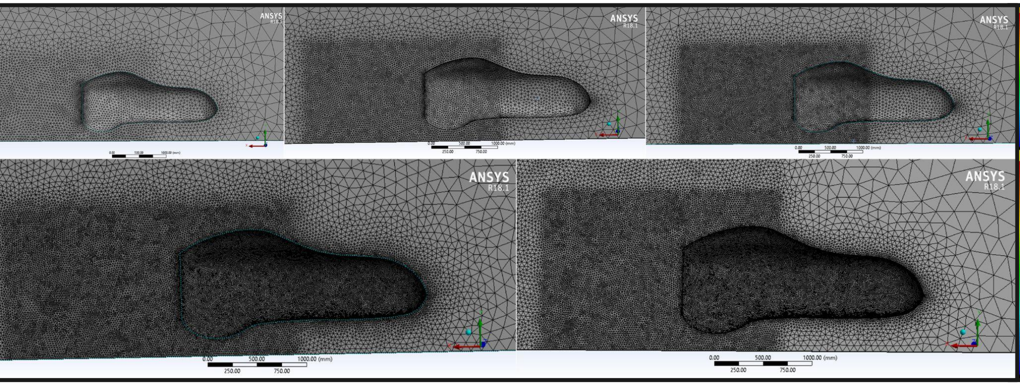


Figure: Different meshes obtained

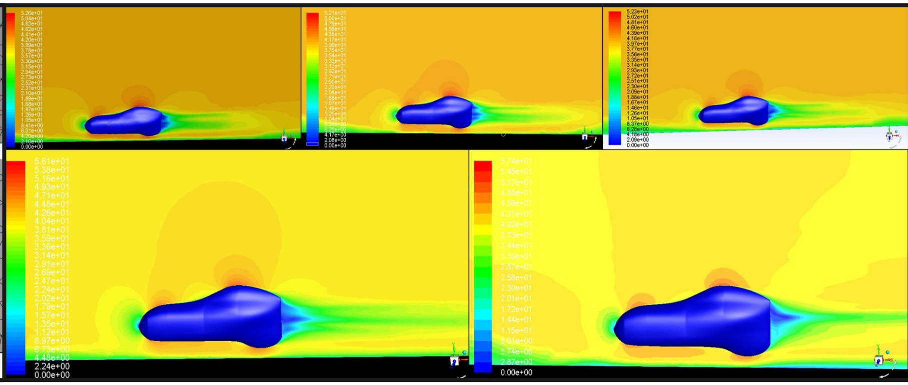
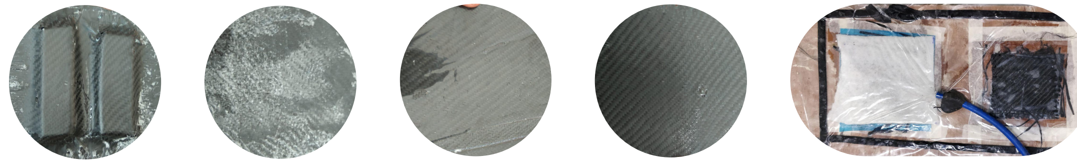


Figure: Velocity Contours of Respective mesh

Design & Development of a Fully Faired Recumbent HPV

Composite Layup & Manufacturing Process



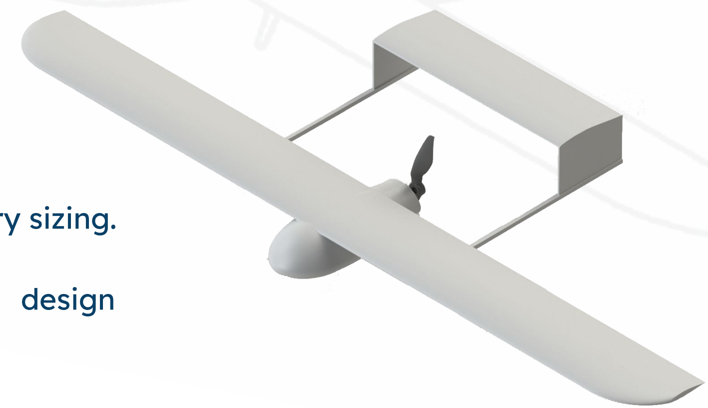
Composite Layup Testing

Manufacturing Process

- Evaluated multiple composite layups using hand layup and resin infusion methods
- Developed fairing molds through sectional patterning and fiberglass tooling
- Assembled carbon fiber fairing panels with controlled curing and surface finish

03 Iterative Design and Aerodynamic Optimization of a Fixed-Wing UAV

- Objective:
  - Develop a fixed-wing UAV optimized for payload capacity and aerodynamic efficiency.
  - Evaluate performance through analytical modeling, CAD design, and CFD validation.
- Design Requirements:
  - Payload Capacity: 5 kg
  - Cruise Velocity: 20 m/s
  - Flight Endurance: 1 hour
  - Target L/D Ratio: ≈ 14
  - Lightweight structure with efficient propulsion and battery sizing.
- Approach:
  - Iterative weight estimation using governing UAV design equations.
  - Aerodynamic optimization using OpenVSP and MATLAB.
  - Final validation using ANSYS CFD.



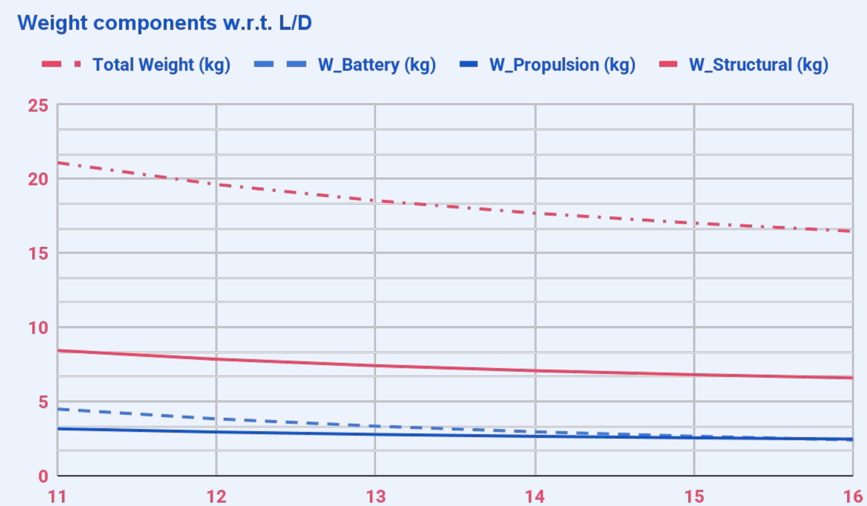
Iterative Weight & Performance Estimation

Step 1: Weight Breakdown Modeling

- Weight Governing Equation:
  - Total Weight (kg) = W_Battery (kg) + W_Propulsion (kg) + W_Structural (kg) + W_Payload (kg)
- Design Assumptions:
  - Structural Weight = 40% of total weight
  - Propulsion Weight = 15% of total weight
  - Battery sizing based on energy requirement:

$$W_{Battery} = \frac{Power \times Time}{Specific\ Energy\ Density}$$

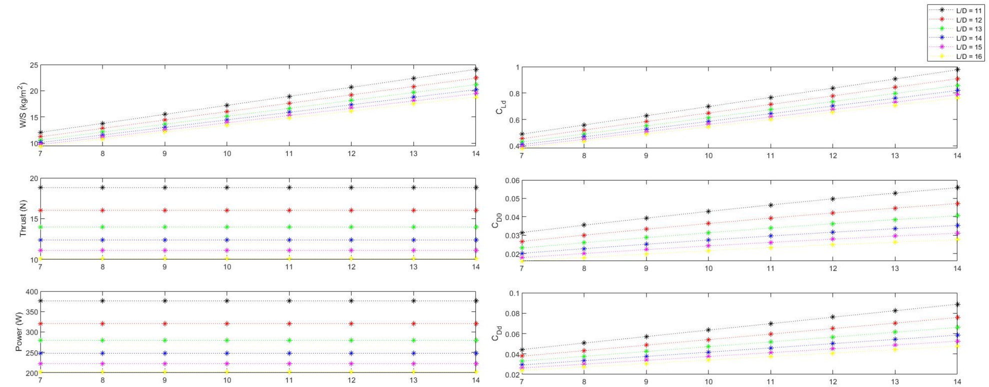
- Key Design Inputs
  - Specific Energy Density = 100 Wh/kg
  - Flight Duration = 1 hour
  - Payload = 5 kg
- Final Selected Design Point:
  - Total Weight ≈ 17.7 kg
  - Power Required ≈ 250 W
  - Thrust Required ≈ 12.4 N



Aerodynamic Design

Step 2: Wing Geometry Optimization

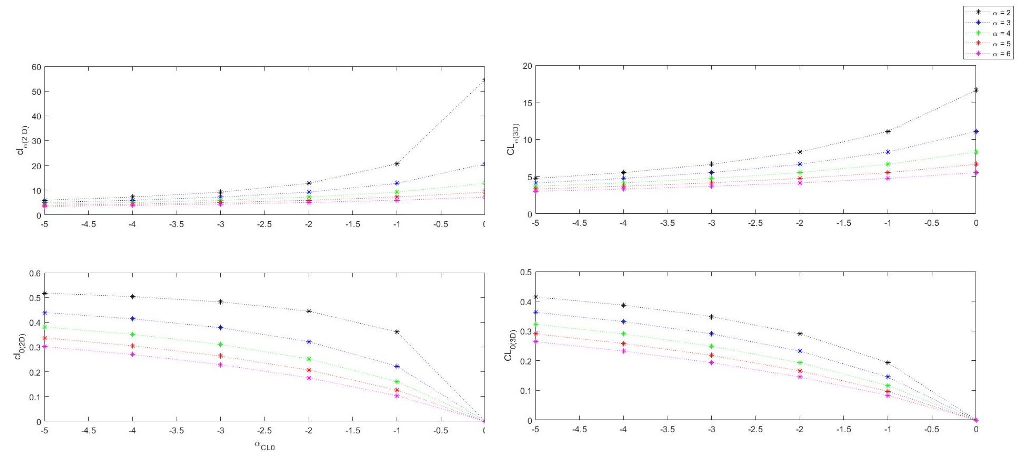
- Final Wing Parameters
  - Wing Area: 1.23 m²
  - Aspect Ratio: 10
  - Lift Coefficient (CL): 0.58
  - Drag Coefficient (CD): 0.042
- Trade-off Considerations
  - Higher AR → Better aerodynamic efficiency
  - Lower AR → Improved maneuverability



Wing Geometry results from MATLAB

Step 3: Airfoil Selection

- Selected Airfoil: NACA 63-412
- Aerodynamic Characteristics
  - CL_0 (3D) = 0.29
  - CLα (3D) = 5.53
  - Design AoA = 3°
- Airfoil chosen for high lift capability and stable cruise performance.
- Tail Geometry & Stability
  - Tail Airfoil: NASA 64-012
  - Tail Area: 0.35 m²
  - CG Location: 0.18 m
  - Trim Angle: 3°



Airfoil Selection results from MATLAB

Iterative Design and Aerodynamic Optimization of a Fixed-Wing UAV

CAD Modeling & CFD Validation

- UAV Design & Simulation
  - Full UAV modeled in SolidWorks
  - Aerodynamic analysis performed in ANSYS Fluent
  - Compared CFD results with OpenVSP predictions.
- Key Engineering Outcomes
  - Achieved required payload capacity.
  - Optimized aerodynamic efficiency.
  - Demonstrated full UAV design workflow:
    - Concept → Analytical modeling → CAD → CFD validation.

CFD Analysis in ANSYS

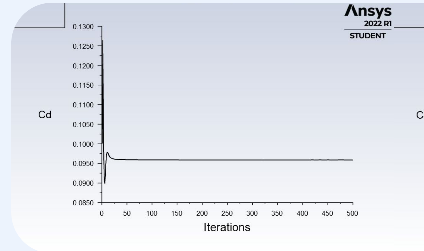


Figure: CD at α = 0

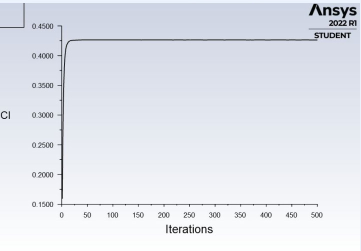
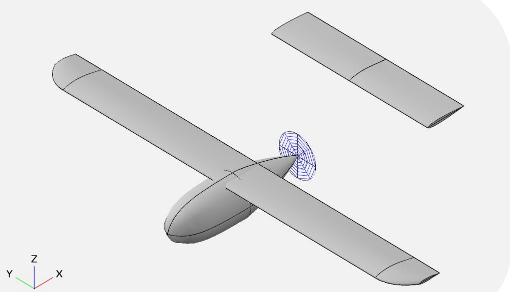


Figure: CD at α = 0

Parameter	CFD Result	Analytical Prediction
CL (α = 0°)	0.42	0.40
CL (α = 3°)	0.75	0.70
CD (α = 3°)	0.12	0.027



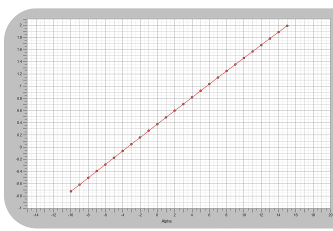


Figure: CL vs α

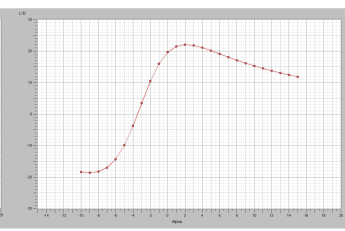


Figure: L/D vs α

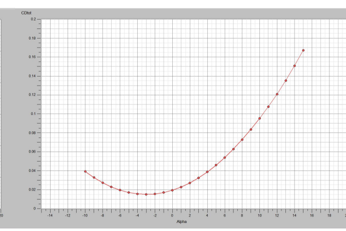


Figure: CD vs α

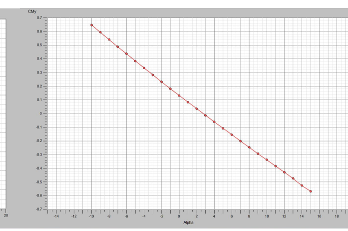
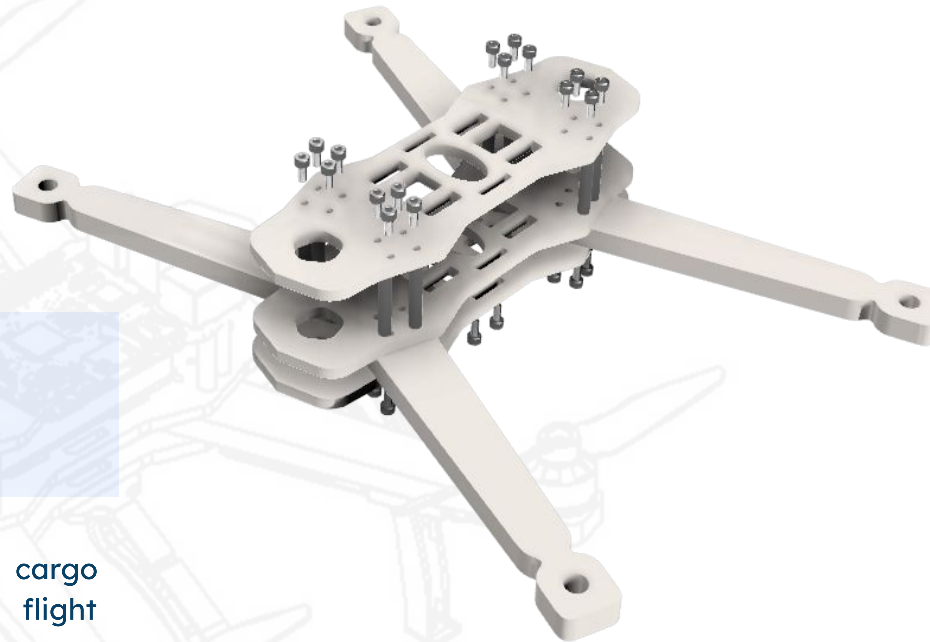


Figure: CM vs α

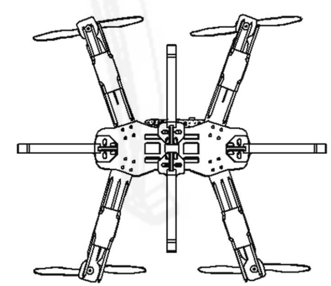
Aerodynamic Analysis results from OpenVSP

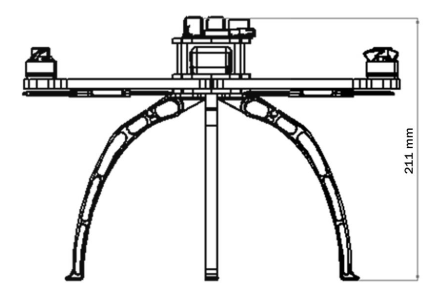


04 Quadcopter Modeling & Structural Optimization

- Modeled and designed an unmanned aerial cargo vehicle to study payload influence on flight performance and environmental sensitivity
- Selected quadcopter configuration enabling 6 DOF control and stable cargo transport capability
- Developed system-level simulation integrating battery performance, propulsion, and vehicle dynamics

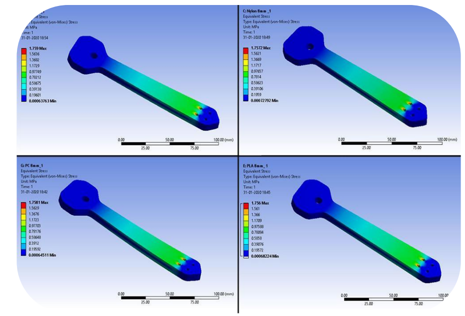
View Published Paper →

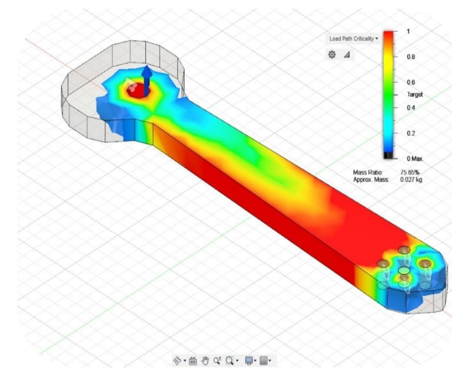




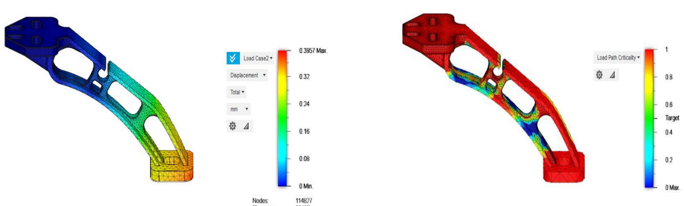
Structural & Mechanical Design

- Designed UAV frame arms (170 mm length) using multi-material evaluation including ABS, PLA, PC, and Nylon PA
- Performed FEA-driven material selection, choosing Nylon for low deformation, manufacturability, and impact resistance
- Applied topology optimization to remove low-stress regions, achieving approximately 25% weight reduction





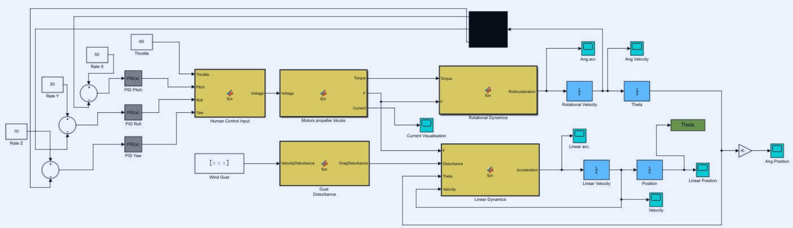
Topology Optimization of UAV Arm



Topology Optimization of UAV Leg

Cargo Handling & Control System Modeling

- Designed servo-actuated pickup-drop mechanism enabling controlled payload release with low power consumption
- Developed integrated PID-based simulation model to analyze UAV motion under varying payload and wind conditions
- Evaluated performance using environmental inputs including 5 m/s wind disturbance and multi-axis rotation rates



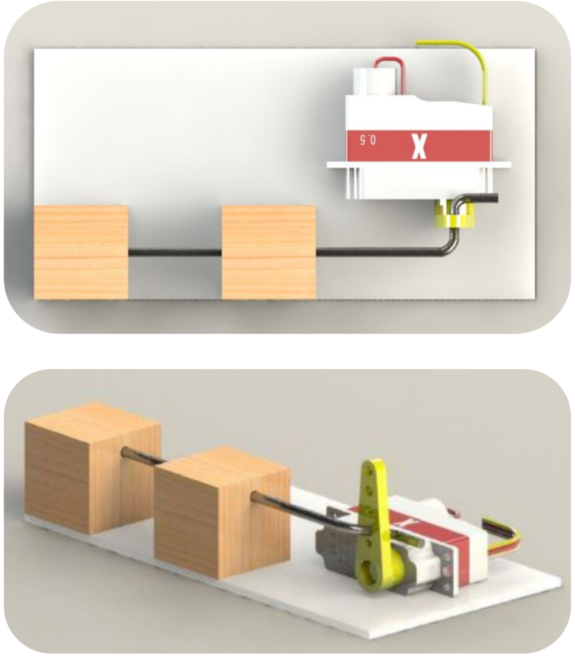


Figure: Pickup-Drop Mechanism

Quadcopter Modeling & Structural Optimization

Validation, Manufacturing & Outcomes

- Validated system performance demonstrating stable flight with payload capacity up to 350 g
- Manufactured structural components using FDM Nylon with 80% honeycomb infill for strength-to-weight optimization
- Demonstrated iterative design methodology improving weight efficiency, payload capability, and delivery feasibility

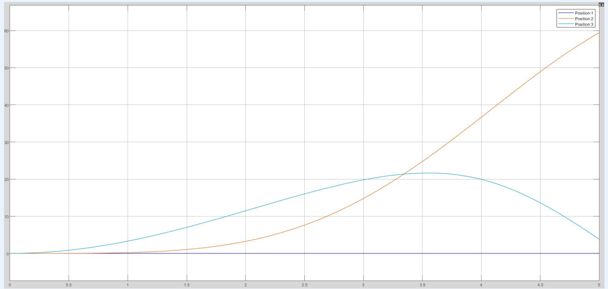


Figure: Linear position of UAV in X, Y and Z axis

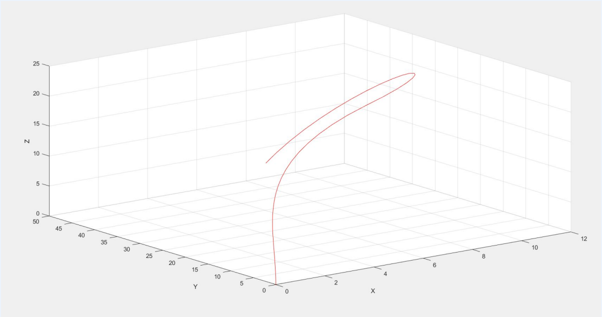


Figure: The path of UAV in 3D space

05 Process Parameter Optimization in Friction Stir Processing

- Investigated the influence of friction stir processing parameters on thermal and mechanical behavior of Aluminium 5083 alloy
- Studied FSP as a solid-state surface modification technique for improving microstructure uniformity and material performance
- Addressed limitations of experimental trials by developing a simulation-based process analysis framework

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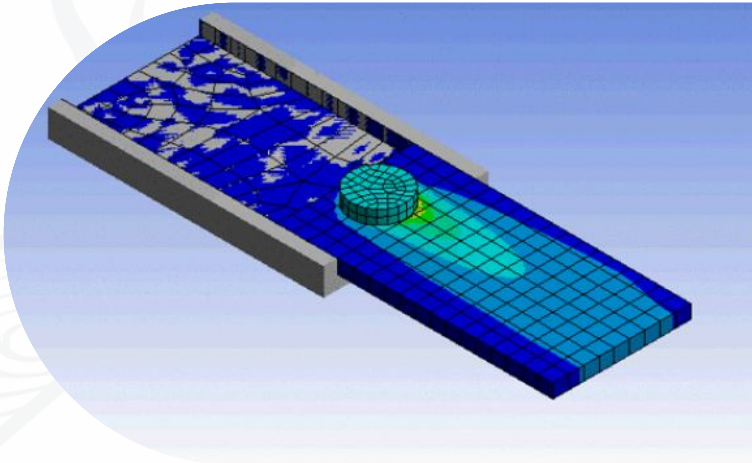


Process Parameter Optimization in Friction Stir Processing

Numerical Modeling & Simulation Methodology

- Developed a 3D thermomechanical finite element model of FSP using ANSYS Workbench 18.1
- Simulated plunging, preheating, and tool traverse stages to evaluate process parameter effects
- Conducted parametric analysis evaluating tool rotational speed, traverse speed, and plunge depth

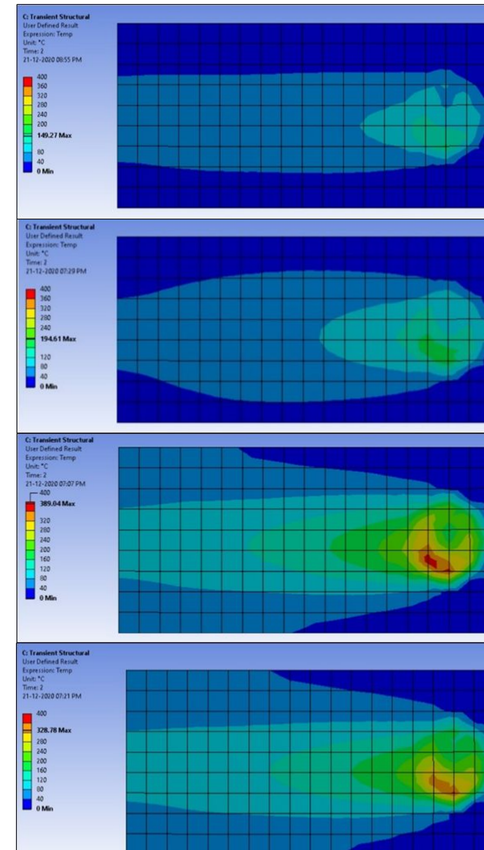
Test No.	Rotational Speed (RPM)	Translational Speed (mm/min)	Plunge Depth (mm)
1	200	60	0.2
2	500	60	0.2
3	200	40	0.2
4	200	60	0.5

Test 1

Test 2

Test 3

Test 4



Results, Validation & Engineering Insights

- Identified strong correlation between process parameters and temperature distribution, with peak temperature reaching 389°C
- Observed increased residual and frictional stresses with higher plunge depth and rotational speeds
- Demonstrated simulation-driven optimization capability to improve weld quality and minimize processing defects

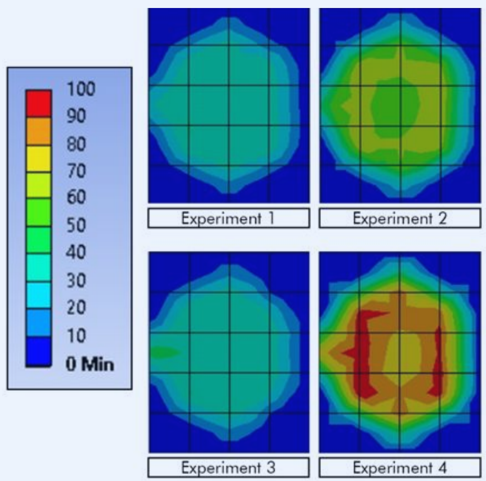


Figure: Analysis of Stress Distribution

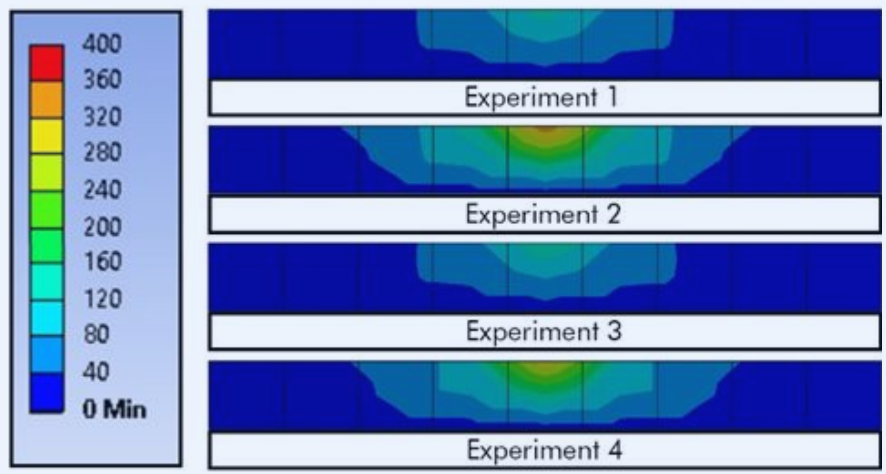


Figure: Analysis of Temperature Distribution

06 Wing Sweep Aerodynamic Study

- Objective:
  - Evaluate impact of sweep angle (λ) on wing aerodynamic performance
- Approach
  - Hess-Smith Panel Method (Python)
  - Boundary Layer Solver Integration
- Parameters Evaluated
  - Cl vs α
  - Cd vs α
  - L/D vs α
  - Boundary Layer Thickness
  - Transition Location
- Tools
  - Python | Numerical Methods | Aerodynamic Modeling

Wing Sweep Aerodynamic Study

Aerodynamic Performance Trends

- Increasing sweep reduces lift curve slope and aerodynamic efficiency
- Minimum drag coefficient increases with higher sweep angles
- Peak L/D achieved at zero sweep, highlighting efficiency trade-offs

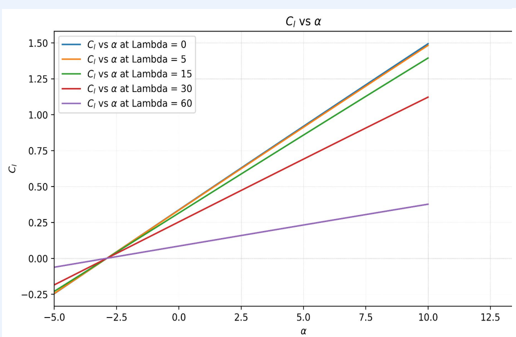


Figure: CL vs α

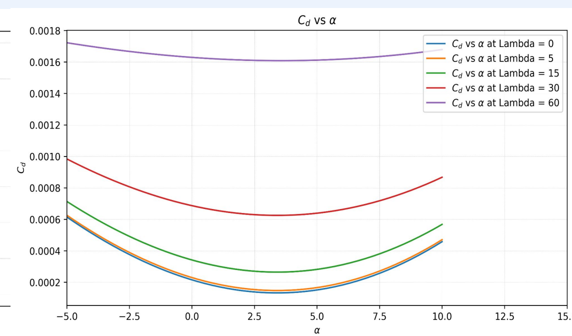


Figure: CD vs α

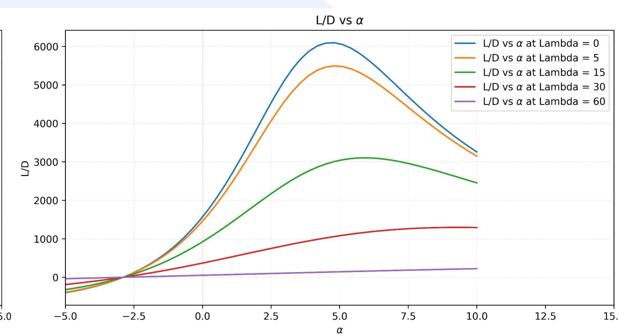


Figure: L/D vs α

Wing Sweep Aerodynamic Study

Flow Physics & Design Insights

- Higher sweep increases boundary layer thickness due to reduced chordwise velocity
- Transition from laminar to turbulent flow delayed at higher sweep angles
- Numerical modeling supports rapid conceptual wing design optimization

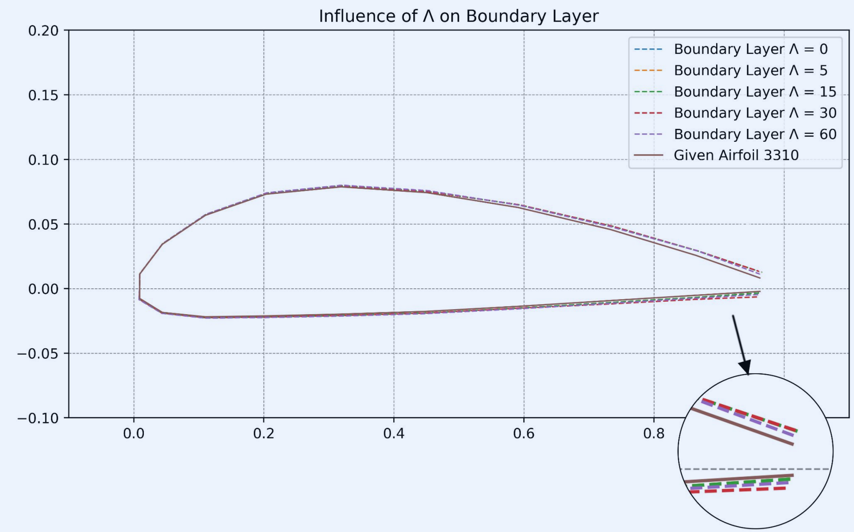


Figure: Influence of sweep angle (λ) on Boundary Layer

07 Beam Impact Test of PVC Pipes

- Evaluated impact resistance of PVC pipe modeled as a simply supported beam
- Integrated axial and transverse strain gauges across span for strain measurement
- Developed full instrumentation system including DAQ modules and controlled loading setup

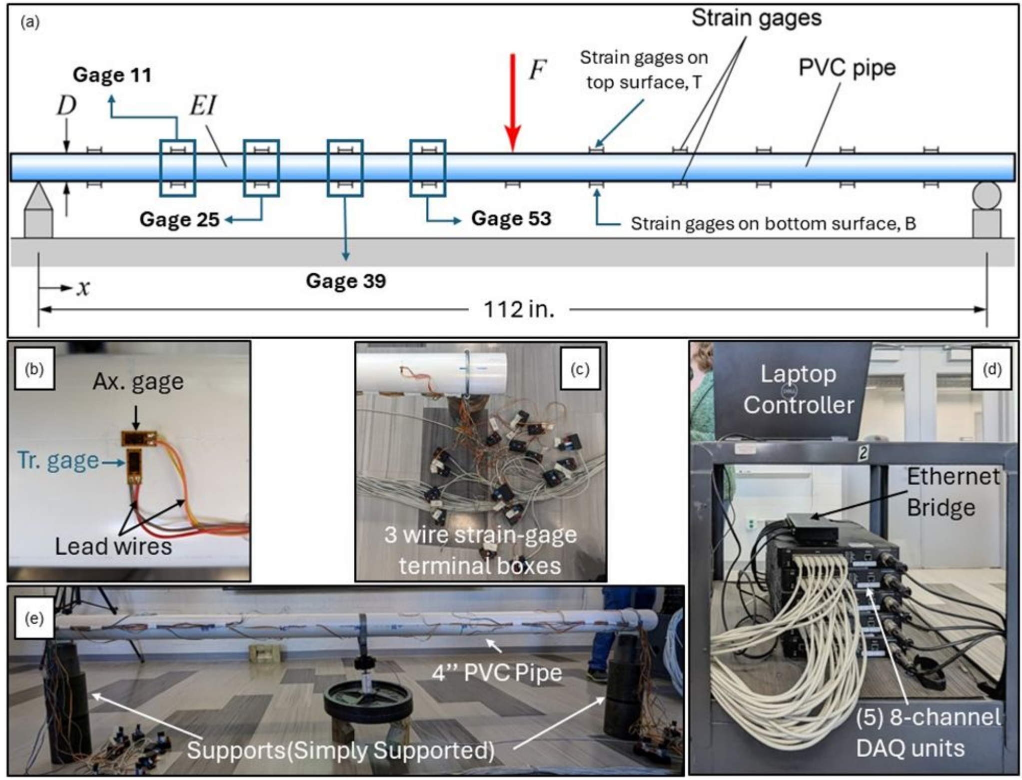


Figure: Experimental Setup

Beam Impact Test of PVC Pipes

Comparative Loading Behavior

- Quasi-static loading produced gradual and predictable strain development
- Dynamic loading generated sharp peaks and oscillatory strain response
- Demonstrated influence of loading rate on structural deformation charac--teristics

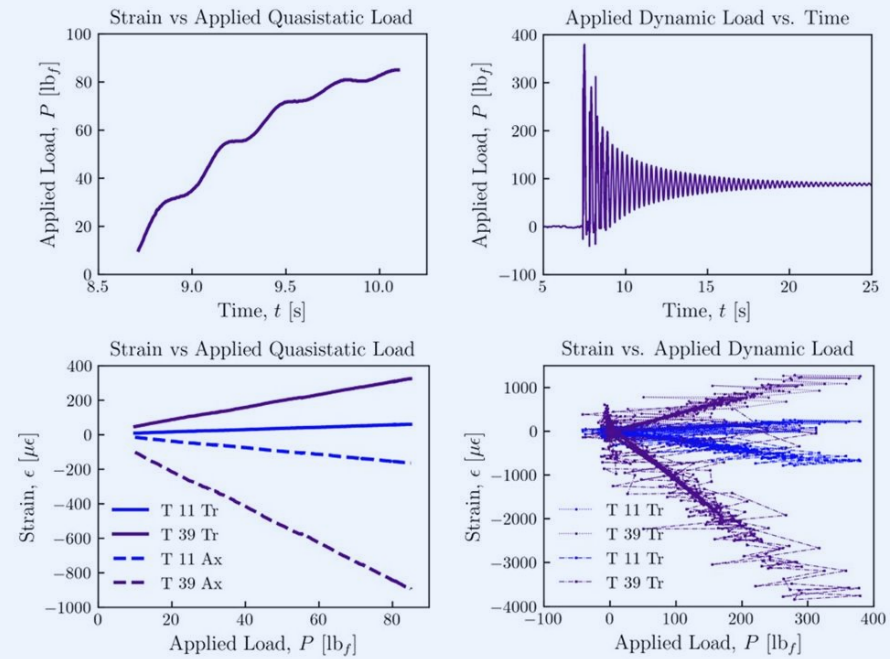


Figure: Quasi-Static Loading

Beam Impact Test of PVC Pipes

Strain Distribution & Material Response

- Strain distribution confirmed peak bending stresses near beam midspan
- Normalized strain trends showed strong elastic recovery and minimal creep
- Load-strain relationship validated elastic response under controlled loading conditions

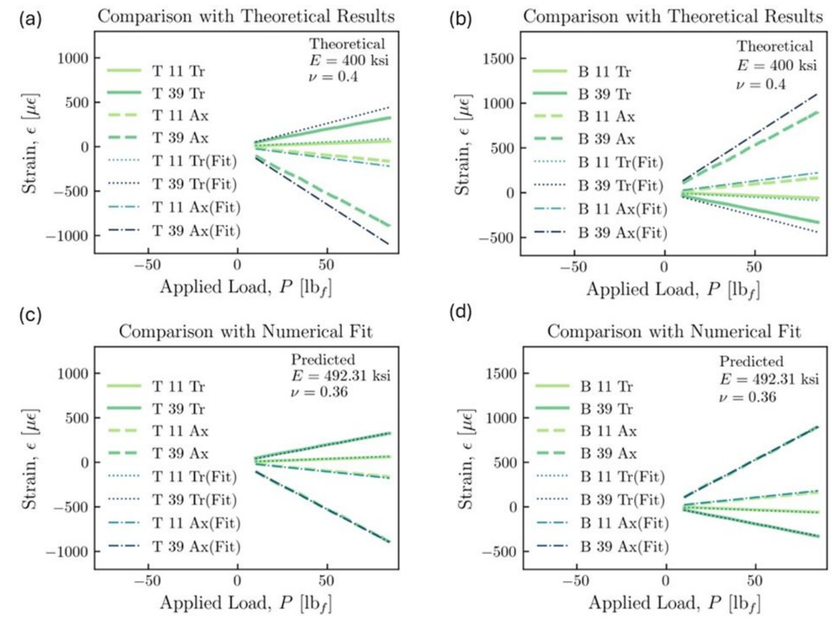


Figure: Strain load analysis

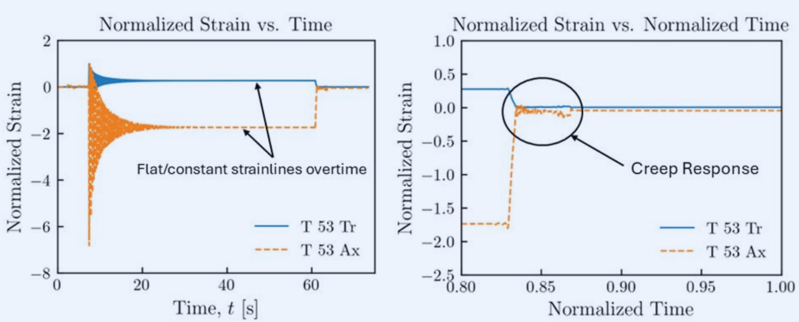


Figure: Normalized Strain Analysis

Beam Impact Test of PVC Pipes

Dynamic Validation & Model Correlation

- Dynamic strain data showed initial agreement with theoretical elastic beam predictions
- Observed deviations highlighted non--linear and localized material effects
- Numerical model calibration improved prediction accuracy using updated material constants

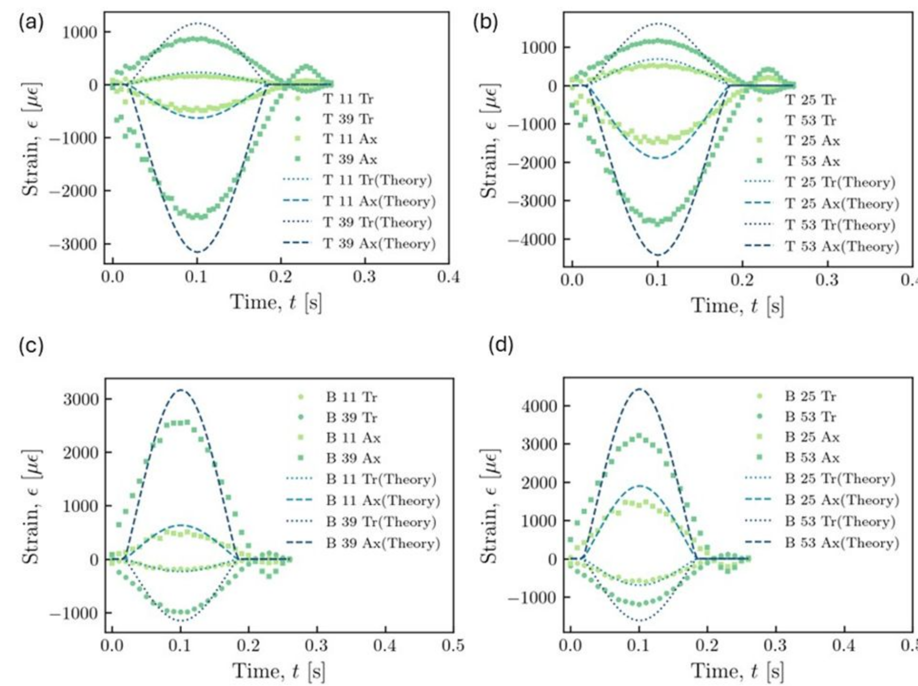


Figure: Dynamic strain analysis

Thanks!

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